MIL-P-38633C(USAF)
17 August 1981
SUPERSEDING
MIL-P-38633B(USAF)
18 December 1968

MILITARY SPECIFICATION

PRIMER, STAB, FOR WDU-4A/A WARHEAD

"This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense."

1. SCOPE

1.1 Scope. This specification establishes the configuration, including constituents, and acceptance requirements of all units of Part Number 66C10112 Primer, Stab, for WDU-4A/A Warhead.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-250	Aluminum-Alloy 2024, Plate and Sheet

MILITARY

MIL-Q-9858	Quality Program Requirements

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1167	Ammunition Data Card

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI, 49016, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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DRAWINGS

Air Force

65B11055	Pin, Firing, Fuze
66C10112	Primer, Fuze

PUBLICATIONS

AFR 71-4	Preparation of hazardous materials for Military Air Shipment
49CFR 71-78	Transportation Interstate Commerce Commission, Explosives and other Dangerous Articles

(Copies of specifications, Standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General Requirements.

3.1.1 Construction and Composition. Construction and composition of the primer shall comply with the detail requirements of this specification.

3.1.2 Interchangeability. The primers manufactured in accordance with this specification shall be completely interchangeable without selection or modification.

3.1.3 First Article Sample. First article primers shall be submitted as designated by the procuring agency, to determine compliance with the requirements and tests of this specification. The first article units shall utilize the manufacturing methods and procedures proposed for the production units and shall be fully interchangeable with them. Further production of the primers by the contractor prior to approval by the procuring agency shall be at the contractor's risk.

3.1.4 Standards of manufacture, manufacturing processes, and production. Parts, materials, finishes, and protective treatments shall comply with the requirements of U.S. Air Force Drawing No. 66C10112 and engineering data assembled thereunder.

3.1.4.1 Substitution. Any substitution shall require written approval prior to use.

3.1.5 Workmanship. Workmanship shall be such as to assure uniform quality and freedom from defects.

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3.2 Functioning. Functioning characteristics defined herein are specified at environmental conditions normal to the manufacturing process and the place of acceptance.

3.2.1 Function and energy output. The primers shall function satisfactorily as evidenced by a depression in the aluminum block within the limits of .030 to .056cm (.012 to .022 inch) when tested as specified in 4.7.3.

3.3 Moisture Content.

3.3.1 Lead Azide. The moisture content of the lead azide at the loading station at the time of loading the primers shall not exceed 0.30 percent when determined as specified in 4.7.4.

3.3.2 Primer Moisture. The moisture content of the primer mixture at the loading station at the time of loading the primer shall not exceed 0.30 percent when determined as specified in 4.7.4.

3.4 Reclaimed Materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended use of this item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Procuring Agency Inspection. The procuring agency shall, at reasonable and prudent times, have access to the contractor's facilities wherein the supplies and services are being prepared.

4.1.2 Procuring agency, Commercial inspection. If a commercial laboratory is employed for any of the tests required, such laboratory shall also provide access to the procuring agency during the course of testing performed on the subject primer.

4.1.3 Procedures. Test procedures shall be submitted to the procuring agency for approval prior to submittal of the first article lot. Test procedures shall be approved by the procuring agency prior to submittal of the lot.

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4.2 Contractor's Quality Assurance. The contractor's quality assurance program shall be in accordance with MIL-Q-9858.

4.3 Classification of inspection. The examination and testing of the primers shall be classified as follows:

a. First article inspection.

b. Acceptance inspection.

4.4 Inspection Conditions. Unless otherwise specified herein, all inspection shall be made under standard room conditions.

4.5 Test equipment. The test equipment shall be similar to that shown in MIL-STD-331 for Test No. 302. The MK 136 MOD 0 Test Set with adapter assembly per LD175382 is suitable for use with this primer.

4.5.1 Approvals. Test equipment shall be approved by the procuring agency prior to use.

4.6 First article inspection. Prior to initiation of regular production, the contractor shall test or submit for test, as designated by the procuring agency, a sample of 160 primers to assure compliance with the requirements of 3.2, inclusive. All parts and materials shall be the same as will be used in production and be obtained from the same source of supply. Prior to tests, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract.

4.6.1 Inspection to be performed. Assemblies shall be subjected by the contractor to the functioning, and energy output tests: One hundred and sixty (160) primer assemblies shall be subjected to test. If a total of four (4) or more assemblies fail to comply with any applicable requirements, the initial inspection sample shall be rejected and the contractor shall take the necessary corrective action and submit a new initial inspection sample of the quantity specified. The functioning and energy output tests shall be performed together.

4.6.2 Witnessing. The procuring agency shall witness any first article tests performed by the contractor. No first article testing shall be conducted without a representative of the procuring agency present throughout the entire test sequence.

4.7 Acceptance Inspections.

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4.7.1 Lot formation. A lot shall mean "inspection lot", i.e., a collection of units of production used on a basis procuring agency inpection. Lots of primers shall consist of units of production over a period of continuous production during which no significant changes in process or formulation of a contractor's raw ingredients have occurred and in which homogeneity within the lot may be expected. The number of primers in a lot shall be as determined by the contract, but shall not exceed 10,000 plus test units, in any instance.

4.7.2 Functioning. A 56.0 gram (2-ounce) steel ball shall be dropped on the firing-pin material, hardness, and point configuration shall be identical to that shown on Air Force Drawing NO. 65B11055 or firing-pin 8797250 shown in Test No. 302 of MIL-STD-331. The remainder of the firing-pin detail may be any practical configuration compatible with the test equipment and consistent with test requirements.

4.7.2.1 Acceptance Criteria. Any primer which fails to function shall be considered defective and the lot shall be judged in accordance with multiple sampling plan of MIL-STD-105 at an AQL of 1.0 percent.

4.7.3 Output test. This test may be conducted at the same time as the functioning test (4.7.2). The maximum depth of indentation produced in each sample disc shall be measured and recorded to the nearest .0025cm (.001 inch). To measure the depth of indentation, mount a dial indicator over a horizontal surface plate in such a manner that the indicator stem is perpendicular to the surface plate. The indicator stem shall be approximately .254cm (0.100 inch) diameter, with a conical tip of 30 degrees included angle terminating in a rounded point having a spherical radius of .0635cm (.025 inch) maximum. Place the disc, indented side up, in a ring type holder so arranged that the periphery of the bottom face of the aluminum disc is parallel to the surface plate. Lower the indicator tip to the bottom of the indentation in the disc and set the dial to zero. Take readings at four equidistant points near the periphery of the upper face of the disc. The arithmetical average of these four measurements shall be considered the depth of indentiation for that disc.

4.7.3.1 Aluminum sample disc. The aluminum sample disc shall have the following characteristics:

a. Diameter 3.098 $\pm$ .025cm (1.22 $\pm$ .01 inches).

b. Thickness .317 $\pm$ .013cm (.125 $\pm$ .005 inches).

c. Material Aluminum, Alloy, Sheet 2024-T3, QQ-A-250/4

4.7.3.2 Acceptance Criteria. Any primer failing to produce an indentation in the aluminum disc within the limits specified in 3.2.1 shall be considered defective. The lot shall be judged in accordance with the Multiple Sampling Plan of MIL-STD-105, at an AQL of 1.0.

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4.7.4 Moisture Content. Moisture test of primer charges shall be conducted at least once for each new batch of material. Moisture content shall be determined on samples of approximately 2 grams of priming mixture or lead azide taken from the operation table immediately prior to consolidation in the cup. Place the sample in a dry, tared container, stopper and weigh the container. Open the container and expose the sample for drying in an oven at 50.0° C ± 15.5° C (122° F ± 4° F) while pulling a partial vacuum of 71.1 cm (28 inches) of mercury, until a constant weight is obtained. Cool the container in a disiccator, stopper and again weigh. Calculate the loss in weight as the percentage of moisture in the sample. All weight shall be accurate to plus or minus 0.2 milligram.

4.7.4.1 Acceptance Criteria. Moisture content exceeding the limits specified in 3.3 shall be cause for the rejection of the batch represented by the sample.

4.8 Primer Data Cards. Data cards in accordance with MIL-STD-1167 shall be provided by the contractor. For each production lot of primers, data cards with all pertinent information recorded shall be prepared. Two (2) data cards shall accompany the shipping invoice, and one (1) shall be placed in each outer packing container

5. PACKAGING

5.1 Preservation - Packaging. Items shall be preserved and packaged to afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use or controlled humidity storage.

5.2 Packing. Items shall be packed to afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. Packing shall conform to ICC Regulation 49 CFR 71.78 and AFR 71.4.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement data shall specify the title, number, and date of this specification.

6.2 Changes from Previous Issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force 99

Preparing Activity
Air Force - 99

Reviewers:
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Replaces edition of 1 Jan 72 which may be used.

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